

Overview of Overfitting machine learning

Subject: Overfitting machine learning

1.

In mathematical modeling, overfitting is the production of an analysis that corresponds too closely or exactly to a particular set of data, and may therefore fail to fit to additional data or predict future observations reliably. An overfitted model is a mathematical model that contains more parameters than can be justified by the data. In the special case of a model that consists of a polynomial function, these parameters represent the degree of a polynomial. The essence of overfitting is to unknowingly extract some of the residual variation (i.e., noise) as if that variation represents the underlying model structure. Underfitting occurs when a mathematical model cannot adequately capture the underlying structure of the data. An under-fitted model is a model that is missing some parameters or terms that would appear in a correctly specified model. Underfitting would occur, for example, when fitting a linear model to nonlinear data. Such a model will tend to have poor predictive performance. The possibility of over-fitting exists when the criterion used for selecting the model is not the same as the criterion used to judge the suitability of a model. For example, a model might be selected by maximizing its performance on some set of training data, yet its suitability might be determined by its ability to perform well on unseen data; overfitting occurs when a model begins to "memorize" training data rather than "learning" to generalize from a trend. As an extreme example, if the number of parameters is the same as or greater than the number of observations, then a model can perfectly predict the training data simply by memorizing the data in its entirety. (For an illustration, see Figure 2.) Such a model will typically fail severely when making predictions. Overfitting is related to both the complexity of the chosen model and how well it is optimized during training. A function class that is too large, in a suitable sense, relative to the dataset size is likely to overfit. Even when the fitted model does not have an excessive number of parameters, it is to be expected that the fitted relationship will appear to perform less well on a new dataset than on the dataset used for fitting (a phenomenon sometimes known as shrinkage). In particular, the value of the coefficient of determination will shrink relative to the original data. To lessen the chance or amount of overfitting, several techniques are available (e.g., model comparison, cross-validation, regularization, early stopping, pruning, Bayesian priors, or dropout). The basis of some techniques is to either (1) explicitly penalize overly complex models or (2) test the model's ability to generalize by evaluating its performance on a set of data not used for training, which is assumed to approximate the typical unseen data that a model will encounter.

2.

Underfitting is the inverse of overfitting, meaning that the statistical model or machine learning algorithm is too simplistic to accurately capture the patterns in the data. A sign of underfitting is that there is a high bias and low variance detected in the current model or algorithm used (the inverse of overfitting: low bias and high variance). This can be gathered from the Bias-variance tradeoff, which is the method of analyzing a model or algorithm for bias error, variance error, and irreducible error. With a high bias and low variance, the result of the model is that it will inaccurately represent the data points and thus insufficiently be able to predict future data results (see Generalization error). As shown in Figure 5, the linear line could not represent all the given data points due to the line not resembling the curvature of the points. One would expect to see a parabola-shaped line as shown in Figure 6 and

Figure 1. If Figure 5 were to be used for analysis, false predictive results would be given contrary to the results if Figure 6 was analyzed. Burnham & Anderson state the following.

3.

Overfitted models ... are often free of bias in the parameter estimators, but have estimated (and actual) sampling variances that are needlessly large (the precision of the estimators is poor, relative to what could have been accomplished with a more parsimonious model). False treatment effects tend to be identified, and false variables are included with overfitted models. ... A best approximating model is achieved by properly balancing the errors of underfitting and overfitting. Overfitting is more likely to be a serious concern when there is little theory available to guide the analysis, in part because then there tend to be a large number of models to select from. The book *Model Selection and Model Averaging* (2008) puts it this way.

4.

Remedy The optimal function usually needs verification on bigger or completely new datasets. There are, however, methods like minimum spanning tree or life-time of correlation that applies the dependence between correlation coefficients and time-series (window width). Whenever the window width is big enough, the correlation coefficients are stable and don't depend on the window width size anymore. Therefore, a correlation matrix can be created by calculating a coefficient of correlation between investigated variables. This matrix can be represented topologically as a complex network where direct and indirect influences between variables are visualized. Dropout regularisation (random removal of training set data) can also improve robustness and therefore reduce over-fitting by probabilistically removing inputs to a layer. Pruning is another technique that mitigates overfitting and enhances generalization by identifying a sparse, optimal neural network structure, while simultaneously reducing the computational cost of both training and inference.

5.

Increase the complexity of the model: If the model is too simple, it may be necessary to increase its complexity by adding more features, increasing the number of parameters, or using a more flexible model. However, this should be done carefully to avoid overfitting. Use a different algorithm: If the current algorithm is not able to capture the patterns in the data, it may be necessary to try a different one. For example, a neural network may be more effective than a linear regression model for some types of data. Increase the amount of training data: If the model is underfitting due to a lack of data, increasing the amount of training data may help. This will allow the model to better capture the underlying patterns in the data. Regularization: Regularization is a technique used to prevent overfitting by adding a penalty term to the loss function that discourages large parameter values. It can also be used to prevent underfitting by controlling the complexity of the model. Ensemble Methods: Ensemble methods combine multiple models to create a more accurate prediction. This can help reduce underfitting by allowing multiple models to work together to capture the underlying patterns in the data. Feature engineering: Feature engineering involves creating new model features from the existing ones that may be more relevant to the problem at hand. This can help improve the accuracy of the model and prevent underfitting.